



INDUSTRIAL ENGINEERING & AUTOMATION PRACTICE

Concept to Commissioning

Industrial Engineering & Automation

EXECUTIVE PRACTICE MANDATES & ADVISORY FOCUS

"The structured 6-stage engineering lifecycle: how custom SPM and automation mandates advance from preliminary concept and DFM review to shop-floor commissioning."

CORE PRACTICE AREAS INCLUDED IN THIS BROCHURE

- **The 6 Stages from Concept to Commissioning**
How Jyot Enterprise manages industrial automation projects with engineering rigor.
- **Stage 1: Requirement Study & Design for Manufacturing**
Preventing costly engineering changes by resolving manufacturing hurdles upfront.
- **Stage 2 & 3: Detailed 3D Modeling, FEA & Electrical Design**
Digital prototyping and comprehensive manufacturing documentation.
- **Stage 4: Precision Manufacturing & Quality Control**
Transforming approved CAD blueprints into physical precision components.

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01 - The 6 Stages from Concept to Commissioning

How Jyot Enterprise manages industrial automation projects with engineering rigor.

Stages 01-03: Design & Engineering Approval

- 01. Requirement Study & DFM Review - Analyzing component drawings, cycle time targets, and process feasibility
- 02. Concept 3D Layout & Simulation - Presenting kinematic 3D layouts, cycle time charts, and commercial proposals
- 03. Detailed Mechanical & Electrical Design - Generating manufacturing drawings, FEA analysis, BOMs, and electrical schematics

Stages 04-06: Build, Test & Factory Commissioning

- 04. Precision Machining & Fabrication - CNC machining, frame fabrication, stress relief, and surface finishing
- 05. Assembly, Wiring & Factory Acceptance Trials (FAT) - Shop floor integration, PLC programming, and 8-hour continuous testing
- 06. Site Installation, Commissioning & Handover - Delivery to customer plant, alignment, utility hookup, operator training, and final sign-off

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02 - Stage 1: Requirement Study & Design for Manufacturing

Preventing costly engineering changes by resolving manufacturing hurdles upfront.

Feasibility Assessment Activities

- Component Geometric Feasibility: Reviewing 2D/3D drawings to identify impossible machining or clamping points
- Cycle Time Budget Breakdown: Slicing the target cycle time into discrete feeding, indexing, clamping, and processing seconds
- Process Risk Identification: Evaluating part burrs, dimensional variations, and material inconsistencies
- Formal Concept Brief Sign-off: Mutually agreed design parameters locking in technical scope

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03 - Stage 2 & 3: Detailed 3D Modeling, FEA & Electrical Design

Digital prototyping and comprehensive manufacturing documentation.

Detailed Engineering Deliverables

- Full Assembly Parametric 3D Model: Complete machine assembly showing all moving mechanisms and stroke limits
- FEA Stress & Deflection Analysis: Verifying critical structural weldments, tooling arms, and bearing housings
- Electrical & Pneumatic Schematics: Complete EPLAN electrical panel diagrams, I/O lists, and pneumatic line circuits
- Indented Bill of Materials (BOM): Specifying material grades, hardened pins, bushings, and commercial buy-out parts

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04 - Stage 4: Precision Manufacturing & Quality Control

Transforming approved CAD blueprints into physical precision components.

Fabrication & Machining Protocols

- Laser Cutting & Certified Welding: Heavy-duty base frames fabricated from IS 2062 structural steel
- Post-Weld Normalization / Stress Relieving: Preventing structural warping and ensuring lifetime geometric stability
- CNC Milling on 3-Axis / 5-Axis Centers: Delivering critical mounting surfaces to +/- 0.010 mm tolerances
- Incoming Quality Inspection (IQC): Verifying all outsourced and in-house parts with calibrated CMM and height gauges

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05 - Stage 5: Shop Floor Assembly & Factory Acceptance Trials

Validating cycle time, repeatability, and machine safety under production conditions.

Testing & Validation Standards

- Mechanical Alignment & Sub-assembly: Meticulous fitting of linear guides, ballscrews, and rotary tables
- Control Panel Wiring: Clean wire ferruling, safety relay wiring, and neat cable track routing
- Dry Cycling & Program Debugging: Verifying sensor debounce, emergency stop safety circuits, and homing routines
- Client Factory Acceptance Test (FAT): Continuous test run on actual components achieving target CPK and cycle time

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06 - Stage 6: Site Commissioning, Training & Steady State

Seamless transition into full production with standing engineering support.

Handover Deliverables

- Customer Plant Rigging & Levelling: Precision anchoring and anti-vibration pad alignment
- Facility Utility Integration: Connecting pneumatic air header and 3-phase electrical drop
- Operator & Maintenance Team Training: Practical hands-on training for daily operation and troubleshooting
- Comprehensive Documentation Pack: Operating manuals, electrical schematics, PLC program backups, and spare parts catalog
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